

Differential Geometry

D. Tong, General Relativity, Problem Sheet 1 (manifolds, tensors, Lie and exterior derivatives).

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General Relativity

Contents

1	Integral Curves of Vector Fields, Q1	1
2	Tensors from Linear Maps, Q2	1
3	The Hessian, Q3	2
4	Transformation of the Metric Determinant, Q4	3
5	Lie Derivatives and Cartan's Magic Formula, Q5	3
6	Properties of the Exterior Derivative, Q6	4
7	Coordinate-Free Definition of the Exterior Derivative, Q7	5
8	Maurer-Cartan Structure Equations on S^3 , Q8	6
9	Coordinate-Induced Bases and the Poincaré Lemma, Q9	6

1. Integral Curves of Vector Fields, Q1

Draw the integral curves corresponding to a vector field in $\mathbb{R}^2$ with components, in Cartesian coordinates, given by

- $X^\mu = (y, -x)$
- $X^\mu = (x - y, x + y)$

[Hint: you might find life easier in polar coordinates.]

Solution. Recall that a vector field can be defined as the tangent to the streamlines at each point, which means

$$X^\mu = \frac{dx^\mu(t)}{dt} \implies \begin{cases} \dot{x} = y, \\ \dot{y} = -x, \end{cases} \quad (1.1)$$

which are solved by the circles on $\mathbb{R}^2$.

For the second vector field, we have

$$\begin{cases} \dot{x} = x - y, \\ \dot{y} = x + y. \end{cases} \quad (1.2)$$

This is solved by spirals of the form

$$x^2 + y^2 = r^2 e^{2 \tan^{-1}(y/x)}. \quad (1.3)$$

2. Tensors from Linear Maps, Q2

Let $\hat{H} : T_p(M) \rightarrow T_p^*(M)$ be a linear map. Define

$$H(X, Y) = [\hat{H}(Y)](X). \quad (2.1)$$

Show that this map is linear in both arguments (e.g. $H(fX + gY, Z) = fH(X, Z) + gH(Y, Z)$ for $f, g \in C^\infty$ and $X, Y, Z \in T_p(M)$) and hence defines a rank $(0, 2)$ tensor. Similarly, show that a linear map $T_p(M) \rightarrow T_p(M)$ defines a tensor of rank $(1, 1)$. What tensor δ arises from the identity map?

Solution. Proof of linearity:

$$H(fX + gY, Z) = [\hat{H}(Z)](fX + gY). \quad (2.2)$$

Since $\hat{H}(Y)$ is a covariant vector, we have

$$H(fX + gY, Z) = f[\hat{H}(Z)](X) + g[\hat{H}(Z)](Y) = fH(X, Z) + gH(Y, Z). \quad (2.3)$$

$$H(X, fY + gZ) = [\hat{H}(fY + gZ)](X). \quad (2.4)$$

By linearity of the linear map $\hat{H}$, we have

$$H(X, fY + gZ) = f[\hat{H}(Y)](X) + g[\hat{H}(Z)](X) = fH(X, Y) + gH(X, Z). \quad (2.5)$$

□

Therefore, H defines a multi-linear map,

$$H : T_p(M) \times T_p(M) \rightarrow \mathbb{R}, \quad (2.6)$$

which is a rank $(0, 2)$ tensor.

Use the linear map,

$$\hat{G} : T_p(M) \rightarrow T_p(M), \quad (2.7)$$

to define

$$G(\omega, X) = \omega(\hat{G}(X)), \quad (2.8)$$

with $\omega \in T_p^*(M)$. Now, we shall check the linearity of the G constructed in both entries:

$$G(f\omega + g\eta, X) = fG(\omega, X) + gG(\eta, X) \quad (2.9)$$

by linearity of covariant vectors, and

$$G(\omega, fX + gY) = fG(\omega, X) + gG(\omega, Y) \quad (2.10)$$

by linearity of $\hat{G}$. Therefore, G is a rank $(1, 1)$ tensor

$$G : T_p^*(M) \times T_p(M) \rightarrow \mathbb{R}. \quad (2.11)$$

The identity map is

$$\hat{\delta}(X) = X. \quad (2.12)$$

Therefore, it defines the tensor

$$\delta(\omega, X) = \omega_\mu X^\mu \quad (2.13)$$

with

$$\delta(f^\mu, e_\nu) = \delta^\mu_\nu. \quad (2.14)$$

3. The Hessian, Q3

Let M be a manifold and $f : M \rightarrow \mathbb{R}$ be a smooth function such that $df = 0$ at some point $p \in M$. Let x^μ be a coordinate chart defined in a neighbourhood of p . Define

$$F_{\mu\nu} = \frac{\partial^2 f}{\partial x^\mu \partial x^\nu}. \quad (3.1)$$

By considering the transformation law for components show that $F_{\mu\nu}$ defines a rank $(0, 2)$ tensor. (This is called the Hessian of f at p .) Construct also a coordinate-free definition and demonstrate its tensorial properties.

Solution. Under the transformation $\tilde{x}^\mu \rightarrow x^\mu$, $F_{\mu\nu}$ transforms as

$$\tilde{F}_{\mu\nu}\Big|_p \rightarrow F_{\mu\nu}\Big|_p = \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial}{\partial x^\rho} \left(\frac{\partial x^\sigma}{\partial \tilde{x}^\nu} \frac{\partial f}{\partial x^\sigma} \right) \Big|_p = \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial x^\sigma}{\partial \tilde{x}^\nu} \frac{\partial^2 f}{\partial x^\rho \partial x^\sigma} \Big|_p + \frac{\partial^2 x^\sigma}{\partial \tilde{x}^\mu \partial \tilde{x}^\nu} \frac{\partial f}{\partial x^\sigma} \Big|_p. \quad (3.2)$$

Given $df|_p = 0$, the second term vanishes, so

$$\tilde{F}_{\mu\nu}\Big|_p \rightarrow \frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial x^\sigma}{\partial \tilde{x}^\nu} F_{\rho\sigma}\Big|_p. \quad (3.3)$$

This is exactly the transformation law of a rank $(0, 2)$ tensor. Therefore, f defines a rank $(0, 2)$ tensor at p .

We can also write the tensor in a coordinate-free way as

$$F(X_p, Y_p) = X(Y(f))|_p = X^\mu Y^\nu \frac{\partial^2 f}{\partial x^\mu \partial x^\nu} \Big|_p + X^\mu \frac{\partial Y^\nu}{\partial x^\mu} \frac{\partial f}{\partial x^\nu} \Big|_p. \quad (3.4)$$

Since $df|_p = 0$, this reduces to

$$F(X_p, Y_p) = X^\mu Y^\nu \frac{\partial^2 f}{\partial x^\mu \partial x^\nu} \Big|_p, \quad (3.5)$$

which is bilinear and symmetric.

4. Transformation of the Metric Determinant, Q4

Let $g_{\mu\nu}$ be a rank $(0, 2)$ tensor. In a basis, one can regard the components $g_{\mu\nu}$ as elements of an $n \times n$ matrix, so that one may define the determinant $g = \det(g_{\mu\nu})$. How does g transform under a change of basis?

Solution. Under a coordinate transformation $x^\mu \rightarrow \tilde{x}^\mu$, we have

$$g \rightarrow \tilde{g} = \det \left(\frac{\partial x^\rho}{\partial \tilde{x}^\mu} \frac{\partial x^\sigma}{\partial \tilde{x}^\nu} g_{\rho\sigma} \right) = \left[\det \left(\frac{\partial x}{\partial \tilde{x}} \right) \right]^2 g. \quad (4.1)$$

5. Lie Derivatives and Cartan's Magic Formula, Q5

Use the Leibniz rule to derive the formula for the Lie derivative of a 1-form ω , valid in any coordinate basis:

$$(\mathcal{L}_X \omega)_\mu = X^\nu \partial_\nu \omega_\mu + \omega_\nu \partial_\mu X^\nu. \quad (5.1)$$

[Hint: consider $(\mathcal{L}_X \omega)(Y)$ for a vector field Y .] Show that the Lie derivative of a $(0, 2)$ tensor g is

$$(\mathcal{L}_X g)_{\mu\nu} = X^\rho \partial_\rho g_{\mu\nu} + g_{\mu\rho} \partial_\nu X^\rho + g_{\rho\nu} \partial_\mu X^\rho. \quad (5.2)$$

For a p -form η , define $\iota_X \eta$ to be the $(p-1)$ -form that results from contracting a vector field X with the first index of η . Show that for a 1-form ω ,

$$\mathcal{L}_X \omega = \iota_X(d\omega) + d(\iota_X \omega). \quad (5.3)$$

Solution. Consider

$$(\mathcal{L}_X \omega)(Y) = \mathcal{L}_X(\omega_\mu dx^\mu)(Y^\nu \partial_\nu) = X^\sigma (\partial_\sigma \omega_\mu) Y^\mu + \omega_\mu \mathcal{L}_X(dx^\mu)(Y^\nu \partial_\nu), \quad (5.4)$$

and use the fact that d and $\mathcal{L}_X$ commute,

$$\mathcal{L}_X(dx^\mu) = d\mathcal{L}_X(x^\mu) = dX^\mu = \partial_\nu X^\mu dx^\nu. \quad (5.5)$$

Therefore, we find

$$(\mathcal{L}_X \omega)(Y) = X^\sigma (\partial_\sigma \omega_\mu) Y^\mu + \omega_\mu (\partial_\sigma X^\mu) Y^\sigma \implies (\mathcal{L}_X \omega)_\mu = X^\nu \partial_\nu \omega_\mu + \omega_\nu \partial_\mu X^\nu. \quad (5.6)$$

For a rank $(0, 2)$ tensor, the Lie derivative is defined as

$$\mathcal{L}_X g = \lim_{t \rightarrow 0} \frac{((\sigma_{-t})_* g)_p - g_p}{t}, \quad (5.7)$$

where the push-forward is given by

$$(\varphi_* g)(X, Y) \Big|_p = g(\varphi_*^{-1} X, \varphi_*^{-1} Y) \Big|_{\varphi^{-1}(p)}, \quad (5.8)$$

and σ_{-t} is the flow generated by $X \in \mathfrak{X}(M)$. The flow equation, in a local coordinate, is given by

$$X^\mu(x) = \frac{dx^\mu}{dt}, \quad (5.9)$$

which is satisfied by

$$x^\mu(t) = x^\mu(0) + tX^\mu(x) + \mathcal{O}(t^2) \quad (5.10)$$

to first order in t (note the slight abuse of notation $x^\mu(\sigma_t) = x^\mu(t)$). In a chart, the push-forward is given by

$$(\varphi_* g)_{\mu\nu} \Big|_p = \frac{\partial x^\sigma}{\partial x^\mu(\varphi)} \frac{\partial x^\rho}{\partial x^\nu(\varphi)} g_{\sigma\rho} \Big|_{\varphi^{-1}(p)}, \quad (5.11)$$

which gives

$$\begin{aligned} ((\sigma_{-t})_* g)_{\mu\nu} \Big|_p &= (\delta_\mu^\sigma - t\partial_\mu X^\sigma)^{-1} (\delta_\nu^\rho - t\partial_\nu X^\rho)^{-1} g_{\sigma\rho} \Big|_{\sigma_t(p)} + \mathcal{O}(t^2) \\ &= (\delta_\mu^\sigma + t\partial_\mu X^\sigma) (\delta_\nu^\rho + t\partial_\nu X^\rho) (g_{\mu\nu} + tX^\lambda \partial_\lambda g_{\mu\nu}) \Big|_p + \dots \\ &= g_{\mu\nu} \Big|_p + t(X^\rho \partial_\rho g_{\mu\nu} + g_{\mu\rho} \partial_\nu X^\rho + g_{\rho\nu} \partial_\mu X^\rho) \Big|_p + \mathcal{O}(t^2), \end{aligned} \quad (5.12)$$

where we have used (5.10) to Taylor expand g around p . Therefore, we find

$$(\mathcal{L}_X g)_{\mu\nu} = X^\rho \partial_\rho g_{\mu\nu} + g_{\mu\rho} \partial_\nu X^\rho + g_{\rho\nu} \partial_\mu X^\rho. \quad (5.13)$$

First, consider

$$\begin{aligned} \iota_X(d\omega) &= \iota_X \frac{1}{2} (\partial_\mu \omega_\nu - \partial_\nu \omega_\mu) dx^\mu \wedge dx^\nu \\ &= \frac{1}{2} (\partial_\mu \omega_\nu - \partial_\nu \omega_\mu) X^\mu dx^\nu - \frac{1}{2} (\partial_\mu \omega_\nu - \partial_\nu \omega_\mu) X^\nu dx^\mu \\ &= X^\mu \partial_\mu \omega_\nu dx^\nu - X^\mu \partial_\nu \omega_\mu dx^\nu. \end{aligned} \quad (5.14)$$

Then,

$$d(\iota_X \omega) = d(\omega_\mu X^\mu) = X^\mu \partial_\nu \omega_\mu dx^\nu + \omega_\mu \partial_\nu X^\mu dx^\nu. \quad (5.15)$$

Therefore, we find

$$\mathcal{L}_X \omega = \iota_X(d\omega) + d(\iota_X \omega). \quad (5.16)$$

6. Properties of the Exterior Derivative, Q6

Let ω be a p -form and η a q -form. Show that the exterior derivative satisfies the properties

- $d(d\omega) = 0$
- $d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p \omega \wedge d\eta$
- $d(\varphi^* \omega) = \varphi^*(d\omega)$ where $\varphi : M \rightarrow N$ for some manifolds M and N

Solution.

$$\begin{aligned} d(d\omega) &= d\left(\frac{1}{p!} \frac{\partial \omega_{\mu_1 \dots \mu_p}}{\partial x^\nu} dx^\nu \wedge dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}\right) \\ &= \frac{1}{p!} \frac{\partial^2 \omega_{\mu_1 \dots \mu_p}}{\partial x^\rho \partial x^\nu} dx^\rho \wedge dx^\nu \wedge dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p} \\ &= 0 \end{aligned} \quad (6.1)$$

by antisymmetry of the wedge product.

$$\begin{aligned} d(\omega \wedge \eta) &= d\left(\frac{1}{p!q!} \omega_{[\mu_1 \dots \mu_p} \eta_{\nu_1 \dots \nu_q]} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p} \wedge dx^{\nu_1} \wedge \dots \wedge dx^{\nu_q}\right) \\ &= \frac{1}{p!q!} ((\partial_{[\mu} \omega_{\mu_1 \dots \mu_p} \eta_{\nu_1 \dots \nu_q]}) + \omega_{[\mu_1 \dots \mu_p} \partial_{[\mu} \eta_{\nu_1 \dots \nu_q]}) dx^\mu \wedge dx^{\mu_1} \wedge \dots \wedge dx^{\nu_q}. \end{aligned} \quad (6.2)$$

Now consider the wedge product

$$\begin{aligned} (d\omega \wedge \eta)_{\mu\mu_1 \dots \nu_q} &= \frac{(p+1)}{(p+1)!q!} (\partial_{[\mu} \omega_{\mu_1 \dots \mu_p} \eta_{\nu_1 \dots \nu_q]}) \\ &= \frac{1}{p!q!} (\partial_{[\mu} \omega_{\mu_1 \dots \mu_p} \eta_{\nu_1 \dots \nu_q]}), \end{aligned} \quad (6.3)$$

which is exactly the first term in (6.2). Similarly, the second term of (6.2) can also be written as $\omega \wedge d\eta$ by permuting μ to the left, which gives a factor of $(-1)^p$. Therefore, we find

$$d(\omega \wedge \eta) = (d\omega) \wedge \eta + (-1)^p \omega \wedge d\eta. \quad (6.4)$$

The pull-back can be written as $\partial \tilde{x}^\mu / \partial x^\nu$. By antisymmetry of the exterior derivative, any term involving $d(\partial \tilde{x}^\mu / \partial x^\nu)$ must vanish. Therefore, the exterior derivative must pass through the pull-back and act on the p -form, ω . That is

$$d(\varphi^* \omega) = \varphi^* d\omega. \quad (6.5)$$

7. Coordinate-Free Definition of the Exterior Derivative, Q7

The exterior derivative of $\omega \in \Lambda^p(M)$ can be defined as

$$\begin{aligned} d\omega(X_1, \dots, X_{p+1}) &= \sum_{j=1}^{p+1} (-1)^{j-1} X_j \left(\omega(X_1, \dots, \hat{X}_j, \dots, X_{p+1}) \right) \\ &\quad + \sum_{j < k} (-1)^{j+k} \omega([X_j, X_k], X_1, \dots, \hat{X}_j, \dots, \hat{X}_k, \dots, X_{p+1}) \end{aligned} \quad (7.1)$$

where a hat denotes omission. In a coordinate basis $\omega = \frac{1}{p!} \omega_{\mu_1 \dots \mu_p} dx^{\mu_1} \wedge \dots \wedge dx^{\mu_p}$, use this definition to determine the components of $d\omega$ in the cases $p = 1$ and $p = 2$.

Solution. For $p = 1$, we have

$$\begin{aligned} d\omega(X, Y) &= X(\omega(Y)) - Y(\omega(X)) - \omega([X, Y]) \\ &= X^\mu \partial_\mu (\omega_\nu Y^\nu) - Y^\nu \partial_\nu (\omega_\mu X^\mu) - \omega_\rho (X^\mu \partial_\mu Y^\rho - Y^\mu \partial_\mu X^\rho) \\ &= (\partial_\mu \omega_\nu - \partial_\nu \omega_\mu) X^\mu Y^\nu, \end{aligned} \quad (7.2)$$

so the components read $(d\omega)_{\mu\nu} = \partial_\mu \omega_\nu - \partial_\nu \omega_\mu = 2 \partial_{[\mu} \omega_{\nu]}$, and

$$d\omega = \partial_\mu \omega_\nu dx^\mu \wedge dx^\nu. \quad (7.3)$$

For $p = 2$, with $\omega = \frac{1}{2} \omega_{\mu\nu} dx^\mu \wedge dx^\nu$, we have $\omega(Y, Z) = \omega_{\mu\nu} Y^\mu Z^\nu$, and

$$\begin{aligned} d\omega(X, Y, Z) &= X(\omega(Y, Z)) - Y(\omega(X, Z)) + Z(\omega(X, Y)) \\ &\quad - \omega([X, Y], Z) + \omega([X, Z], Y) - \omega([Y, Z], X). \end{aligned} \quad (7.4)$$

Expanding the first piece,

$$X(\omega(Y, Z)) = X^\rho(\partial_\rho\omega_{\mu\nu})Y^\mu Z^\nu + \omega_{\mu\nu}X^\rho(\partial_\rho Y^\mu)Z^\nu + \omega_{\mu\nu}X^\rho Y^\mu(\partial_\rho Z^\nu), \quad (7.5)$$

and similarly for the other two, while each commutator contributes terms such as

$$\omega([X, Y], Z) = \omega_{\rho\nu}(X^\sigma\partial_\sigma Y^\rho - Y^\sigma\partial_\sigma X^\rho)Z^\nu. \quad (7.6)$$

All terms involving derivatives of the vector fields cancel pairwise after relabeling dummy indices and using antisymmetry of $\omega_{\mu\nu}$. What remains is

$$d\omega(X, Y, Z) = (\partial_\rho\omega_{\mu\nu} + \partial_\mu\omega_{\nu\rho} + \partial_\nu\omega_{\rho\mu})X^\rho Y^\mu Z^\nu, \quad (7.7)$$

so the components read

$$(d\omega)_{\rho\mu\nu} = \partial_\rho\omega_{\mu\nu} + \partial_\mu\omega_{\nu\rho} + \partial_\nu\omega_{\rho\mu} = 3\partial_{[\rho}\omega_{\mu\nu]}, \quad (7.8)$$

which gives

$$d\omega = \frac{1}{2}\partial_\rho\omega_{\mu\nu}dx^\rho \wedge dx^\mu \wedge dx^\nu. \quad (7.9)$$

Both cases fit the general pattern $(d\omega)_{\mu_0\mu_1\cdots\mu_p} = (p+1)\partial_{[\mu_0}\omega_{\mu_1\cdots\mu_p]}$.

8. Maurer-Cartan Structure Equations on S^3 , Q8

A three-sphere can be parameterized by Euler angles (θ, ϕ, ψ) where $0 < \theta < \pi$, $0 < \phi < 2\pi$, $0 < \psi < 4\pi$. Define the following 1-forms

$$\begin{aligned} \sigma^1 &= -\sin\psi d\theta + \cos\psi \sin\theta d\phi, \\ \sigma^2 &= \cos\psi d\theta + \sin\psi \sin\theta d\phi, \\ \sigma^3 &= d\psi + \cos\theta d\phi. \end{aligned}$$

Show that $d\sigma^1 = \sigma^2 \wedge \sigma^3$ with analogous results for $d\sigma^2$ and $d\sigma^3$.

Solution.

$$d\sigma^1 = -\cos\theta d\psi \wedge d\theta - \sin\psi \sin\theta d\psi \wedge d\phi + \cos\psi \cos\theta d\theta \wedge d\phi, \quad (8.1)$$

and

$$\begin{aligned} \sigma^2 \wedge \sigma^3 &= (\cos\psi d\theta + \sin\psi \sin\theta d\phi) \wedge (d\psi + \cos\theta d\phi) \\ &= \cos\psi d\theta \wedge d\psi + \sin\psi \sin\theta d\phi \wedge d\psi + \cos\psi \cos\theta d\theta \wedge d\phi, \end{aligned} \quad (8.2)$$

which clearly shows

$$d\sigma^1 = \sigma^2 \wedge \sigma^3. \quad (8.3)$$

Similarly, one can obtain analogous results for $d\sigma^2$ and $d\sigma^3$.

9. Coordinate-Induced Bases and the Poincaré Lemma, Q9

Let $\{e_\mu\}$ be a basis of vector fields, with

$$[e_\mu, e_\nu] = \gamma^\rho_{\mu\nu} e_\rho. \quad (9.1)$$

The functions $\gamma^\rho_{\mu\nu}$ are known as commutator components. For a choice of coordinates $\{x^\mu\}$, we often work with the coordinate induced basis $e_\mu = \{\partial_\mu\}$. Show that, in this case, $[e_\mu, e_\nu] = 0$.

The purpose of this question is to show the converse: that $[e_\mu, e_\nu] = 0$ only for a coordinate induced basis. Consider a general basis $\{e_\mu\}$ and the dual basis $\{f^\mu\}$ of one-forms. In general, these can be expanded as

$$e_\mu = e_\mu^\rho \frac{\partial}{\partial x^\rho} \quad \text{and} \quad f^\mu = f^\mu_\rho dx^\rho \quad (9.2)$$

where $e_\mu{}^\rho f^\nu{}_\rho = \delta_\nu^\mu$. Show that

$$e_\mu{}^\sigma \frac{\partial e_\nu{}^\lambda}{\partial x^\sigma} - e_\nu{}^\sigma \frac{\partial e_\mu{}^\lambda}{\partial x^\sigma} = \gamma^\rho{}_{\mu\nu} e_\rho{}^\lambda. \quad (9.3)$$

Hence deduce that

$$e_\mu{}^\sigma e_\nu{}^\lambda \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} - e_\nu{}^\sigma e_\mu{}^\lambda \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} = -\gamma^\rho{}_{\mu\nu}, \quad (9.4)$$

and finally that

$$\frac{\partial f^\rho{}_\sigma}{\partial x^\lambda} - \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} = -\gamma^\rho{}_{\mu\nu} f^\mu{}_\lambda f^\nu{}_\sigma. \quad (9.5)$$

Use this result, together with the Poincaré lemma, to show that if $[e_\mu, e_\nu] = 0 \forall \mu, \nu$ then the basis is coordinate induced.

Solution. For coordinate bases, we clearly have

$$[\partial_\mu, \partial_\nu] = 0. \quad (9.6)$$

Now, consider the commutation relation

$$[e_\mu{}^\rho \partial_\rho, e_\nu{}^\sigma \partial_\sigma] = \gamma^\lambda{}_{\mu\nu} e_\lambda{}^\delta \partial_\delta. \quad (9.7)$$

The left-hand side is

$$[e_\mu{}^\rho \partial_\rho, e_\nu{}^\sigma \partial_\sigma] = e_\mu{}^\rho (\partial_\rho e_\nu{}^\sigma) \partial_\sigma - e_\nu{}^\sigma (\partial_\sigma e_\mu{}^\rho) \partial_\rho. \quad (9.8)$$

After relabeling, we find

$$e_\mu{}^\sigma \frac{\partial e_\nu{}^\lambda}{\partial x^\sigma} - e_\nu{}^\sigma \frac{\partial e_\mu{}^\lambda}{\partial x^\sigma} = \gamma^\rho{}_{\mu\nu} e_\rho{}^\lambda. \quad (9.9)$$

Then, consider the expression

$$e_\mu{}^\sigma \frac{\partial (e_\nu{}^\lambda f^\rho{}_\lambda)}{\partial x^\sigma} - e_\nu{}^\sigma \frac{\partial (e_\mu{}^\lambda f^\rho{}_\lambda)}{\partial x^\sigma}. \quad (9.10)$$

Since $e_\mu{}^\rho f^\nu{}_\rho = \delta_\nu^\mu$, the above expression must vanish. Therefore, we must have

$$\begin{aligned} 0 &= e_\mu{}^\sigma \frac{\partial (e_\nu{}^\lambda f^\rho{}_\lambda)}{\partial x^\sigma} - e_\nu{}^\sigma \frac{\partial (e_\mu{}^\lambda f^\rho{}_\lambda)}{\partial x^\sigma} \\ &= e_\mu{}^\sigma f^\rho{}_\lambda \frac{\partial e_\nu{}^\lambda}{\partial x^\sigma} - e_\nu{}^\sigma f^\rho{}_\lambda \frac{\partial e_\mu{}^\lambda}{\partial x^\sigma} + e_\mu{}^\sigma e_\nu{}^\lambda \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} - e_\nu{}^\sigma e_\mu{}^\lambda \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} \\ &= \gamma^\rho{}_{\mu\nu} + e_\mu{}^\sigma e_\nu{}^\lambda \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} - e_\nu{}^\sigma e_\mu{}^\lambda \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma}. \end{aligned} \quad (9.11)$$

Therefore, we find

$$e_\mu{}^\sigma e_\nu{}^\lambda \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} - e_\nu{}^\sigma e_\mu{}^\lambda \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} = -\gamma^\rho{}_{\mu\nu}. \quad (9.12)$$

Then, contracting with f , we find

$$\frac{\partial f^\rho{}_\sigma}{\partial x^\lambda} - \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} = -\gamma^\rho{}_{\mu\nu} f^\mu{}_\lambda f^\nu{}_\sigma. \quad (9.13)$$

The Poincaré lemma states that, on $M = \mathbb{R}^n$, closed implies exact. Multiplying the previous expression by $dx^\lambda \wedge dx^\sigma$ and using antisymmetry of the wedge,

$$d\theta^\rho = \frac{1}{2} \left(\frac{\partial f^\rho{}_\sigma}{\partial x^\lambda} - \frac{\partial f^\rho{}_\lambda}{\partial x^\sigma} \right) dx^\lambda \wedge dx^\sigma = -\frac{1}{2} \gamma^\rho{}_{\mu\nu} \theta^\mu \wedge \theta^\nu, \quad (9.14)$$

where $\theta^\rho = f^\rho{}_\sigma dx^\sigma$ is the dual basis to $e_\mu = e_\mu{}^\rho \partial_\rho$. Now suppose $\gamma^\rho{}_{\mu\nu}$ vanish. Then $d\theta^\rho = 0$, so θ^ρ is a closed 1-form. By the Poincaré lemma, on $M = \mathbb{R}^n$ (or any contractible neighbourhood) closed forms are exact, so there exist functions y^ρ with

$$\theta^\rho = dy^\rho. \quad (9.15)$$

The y^ρ are functionally independent because θ^ρ are linearly independent at every point, so they form a coordinate system. In these coordinates, $\theta^\rho(e_\mu) = \delta_\mu^\rho$ becomes $\partial y^\rho / \partial y^\mu_{(e)} = \delta_\mu^\rho$, i.e. $e_\mu = \partial / \partial y^\mu$, a coordinate basis.

Conversely, if $e_\mu = \partial / \partial y^\mu$ for some coordinates y^μ , then $[e_\mu, e_\nu] = [\partial_{y^\mu}, \partial_{y^\nu}] = 0$, so $\gamma^\rho{}_{\mu\nu} = 0$. Therefore $\{e_\mu\}$ is (locally) a coordinate basis if and only if $\gamma^\rho{}_{\mu\nu} = 0$.